

Department of Computer Science and Engineering

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Project Title: Mesh-Search: A Privacy-First 3D Geometric AI Framework for Missing Person Identification (8/6/2026)

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Introduction

- **Traditional processes often rely on manual review of reports and CCTV footage**, which can be slow and labor-intensive—especially during large-scale searches.
- This system uses computer vision and face recognition to compare faces detected in surveillance frames with the stored images of registered missing cases.
- When a match is found, the system automatically triggers an alert (email notification) to the concerned family/police contacts, reducing response time and improving the chances of reunion.

Key components in this project:

- **Missing case registration** (details + image stored in database)
- **Surveillance mode** (browser webcam frames → backend processing)
- **Face recognition matching** using face_recognition encodings and a threshold decision
- **Cooldown-protected notification alerts** to avoid repeated spamming

Outcome: faster detection, lower manual workload, and timely alerts that support effective search and rescue.

Problem statement

Missing persons represent an urgent public safety concern where time plays a crucial role.

In the current system, identifying whether a person seen in surveillance footage matches a missing-case record is often done manually.

Key challenges in the existing process:

- Manual CCTV review is slow and labor-intensive, especially across multiple cameras and long durations.
- Authorities and family members rely on manual cross-checking of reports and images, which delays decision-making.
- Matching missing-case information to real-time/surveillance footage is error-prone due to differences in lighting, angle, and image quality.
- Repeated alerts may occur without automated cooldown/notification control, leading to notification fatigue.

Objectives

- **Automated Case Registration and Management:** To develop a secure, centralized web platform using the Django framework for managing missing person records, enabling the storage of biometric photographs alongside critical metadata like Aadhar details and contact information.
- **Real-Time Biometric Surveillance and Identification:** To implement an intelligent monitoring module using OpenCV and deep learning-based facial recognition (128-d embeddings) to scan live camera feeds and accurately identify missing individuals in unconstrained public environments.
- **Instant Automated Alert and Recovery System:** To engineer a rapid notification engine that triggers automated email and mobile alerts the moment a match is detected, providing stakeholders with real-time sighting data, timestamps, and locations to facilitate immediate rescue.

Literature survey

No.	Author / Year	Title of Paper	Method / Technique	Key Contribution
1.	K. Ayappan, 2020	Facial recognition for Lost Persons using ML	Machine learning + facial recognition	Automated identification of missing persons using face matching
2	R. Geetha et al., 2020	A Review of Missing Person Detection Using Facial Recognition Techniques	Survey of facial recognition methods	Reviews algorithms, datasets, and performance metrics
3	A Chavda et al.,	Classroom Behavior Detection with Deformable DETR and Swin Transformer	Transformer-based detection	Combined Swin Transformer and Deformable DETR to detect student actions and analyze classroom interactions.
4	Jia et al., 2024	Student Behavior Recognition in Classroom	YOLOv5 + OpenPose	Proposed a deep learning model combining YOLOv5 and pose estimation to detect student activities and engagement.
5	Mahmood et al., 2024	Measuring Student Engagement through Behavioral and Emotional Features	Deep learning	Developed a framework to analyze both emotional and behavioral engagement of students during lectures.

Software requirement specifications

Functional Requirements

The system should allow registration of missing person details and face images.

The system should capture and process real-time webcam/surveillance frames.

The system should detect, encode, and match faces using facial recognition.

The system should generate automated email alerts on successful matches.

The system should display detection results and support admin CRUD operations.

Non-Functional Requirements

The system should process frames in near real-time with reliable accuracy.

The system should ensure secure storage of personal data and images.

The system should handle invalid frames or no-face cases without crashing.

The system should support multiple missing cases and surveillance sessions.

The system should provide a user-friendly surveillance interface.

SRS Contd...

Hardware Requirements

Webcam/camera for surveillance mode.

Minimum 8 GB RAM and multi-core processor recommended.

Sufficient storage for uploaded images and database records.

Software Requirements

Operating System: Windows / Linux / macOS

Backend Framework: Django

Libraries: OpenCV, face_recognition, NumPy

Database: SQLite

Frontend: HTML, CSS, JavaScript, Bootstrap

Notification System: SMTP Email Service

Detailed Functional requirements

Missing Person Registration

The system should allow users to register missing person details along with facial images. The uploaded information should be stored securely in the database for future matching and retrieval.

Real-Time Surveillance Monitoring

The system should capture real-time video frames from webcams or surveillance cameras. The captured frames should be continuously processed for face detection and recognition.

Face Detection and Recognition

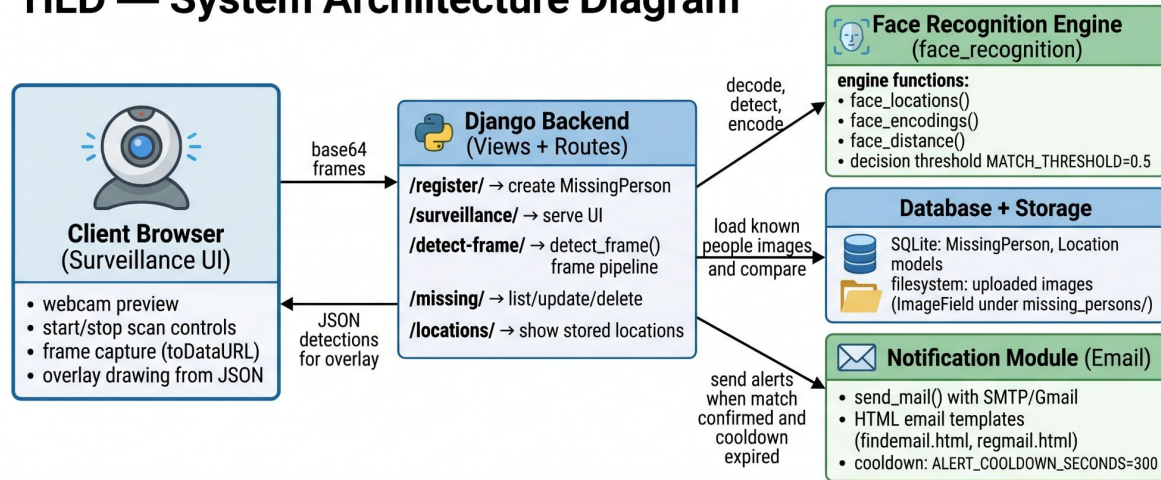
The system should detect human faces from input frames and generate facial encodings using computer vision techniques. The detected faces should be matched with registered missing person records.

Alert and Notification Generation

The system should generate automated alerts when a matching face is identified. Notifications containing detected person details should be sent to concerned authorities or family members through email services.

System design

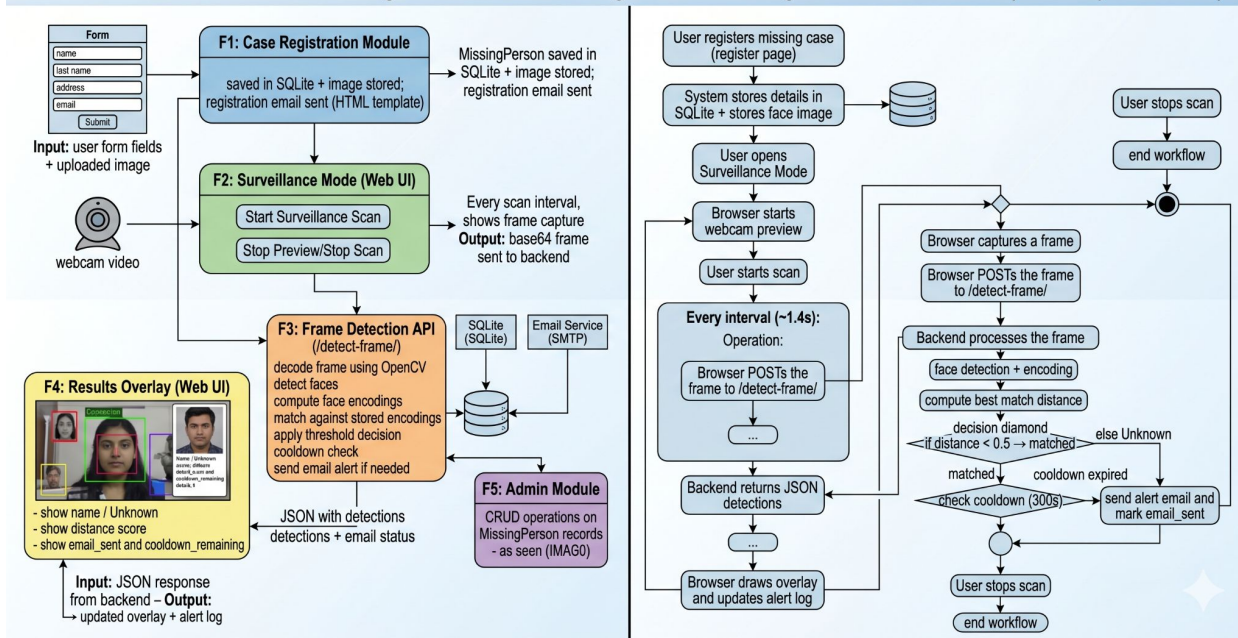
Bharatiya Rescue Missing Person Detection HLD — System Architecture Diagram



“Bharatiya Rescue uses a Django backend to process webcam frames, match detected faces against registered missing cases in SQLite, and send cooldown-protected email alerts.”

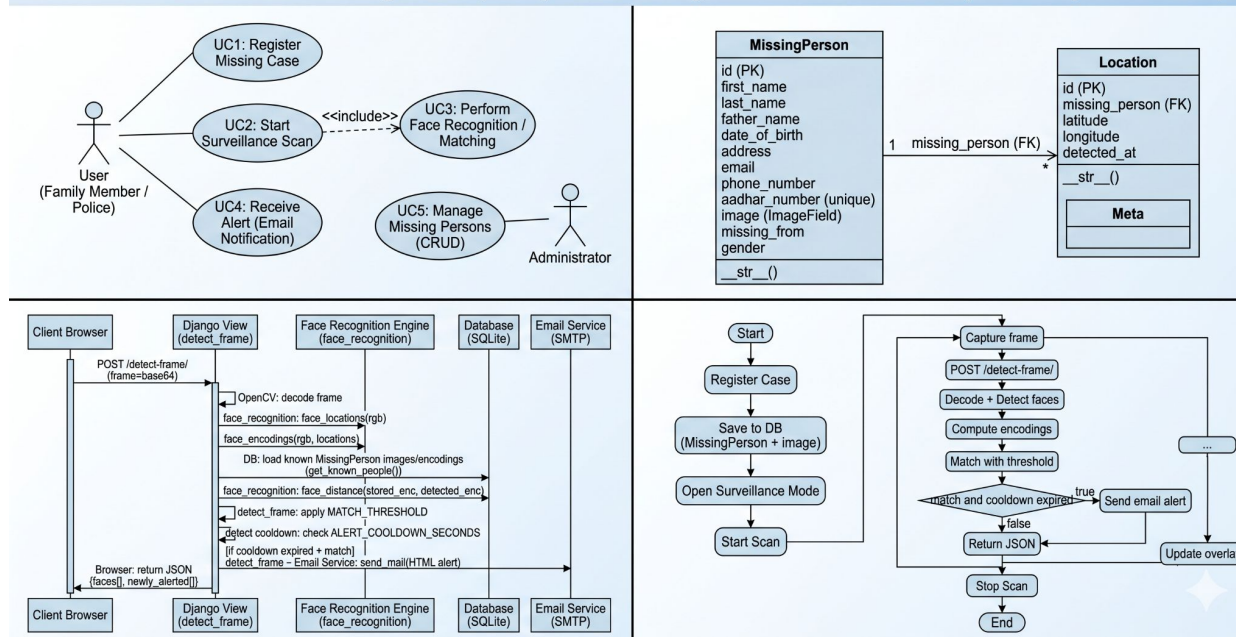
Functional diagram

Slide Set 6.2 — Functional Diagram / Workflow Diagram — Missing Person Detection (Bharatiya Rescue)



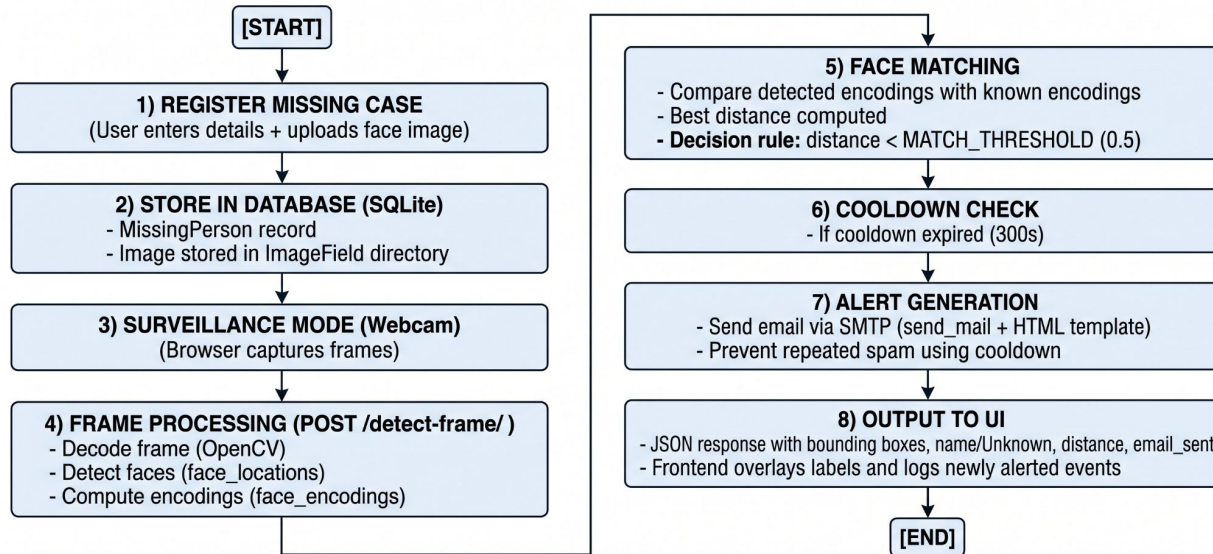
UML Diagram

Slide Set 6.5 — UML Diagrams (LLD) — Missing Person Detection (Bharatiya Rescue)



Proposed Diagram

Proposed Concept — “Register → Detect → Match → Alert”



Bharatiya Rescue automates missing-person identification by matching surveillance faces with registered cases and triggering cooldown-protected email alerts.

Implementation- Algorithm

1. Start
2. Initialize webcam surveillance system
3. Capture real-time video stream
4. WHILE surveillance scan is active DO
5. Extract video frames from webcam stream
6. Preprocess frame (decode and convert BGR to RGB)
7. Detect faces in the frame
8. FOR each detected face DO
9. Generate face encoding using facial recognition model
10. Compare encoding with registered missing person database
11. Compute similarity distance and identify best match
12. IF distance < threshold THEN
13. Mark detected person as matched
14. Send alert email notification
15. ELSE
16. Mark detected face as unknown
17. END IF
18. END FOR
19. Display detection results with bounding boxes and labels
20. END WHILE
21. Stop surveillance scan
22. Stop.

Testing

Test Case ID	Module	Test Scenario	Input	Expected Output	Result
TC01	Home Module	Check homepage navigation links	User click actions	Correct pages opened successfully	Pass/Fail
TC02	Face Detection	Detect student faces from frame	Classroom image/video frame	All visible faces detected with bounding boxes	Pass/Fail
TC03	Face Recognition	Recognize registered student	Known student face	Student name/ID displayed correctly	Pass/Fail
TC04	Missing Case Registration Module	Register missing person details	Form data + image upload	Missing person record stored successfully	Pass/Fail

Results Snapshots

Report Missing Case

First Name	Last Name
Enter Your First Name	Enter Your Surname
Father's Name	Date of Birth
Enter Father's Name	mm/dd/yyyy
Address	
Enter your Address	
Email	Phone Number
your@email.com	Phone Number
We'll never share your email with anyone else.	
Aadhar Number	Missing Date
XXXX XXXX XXXX	mm/dd/yyyy

Welcome to Survivala

Start Webcam

Press q to Close Camera.



Establishing secure connection...

Comparative Analysis

Parameter	Existing ML-Based Systems	Proposed Missing Person Detection System
Face Recognition Accuracy	80–90% accuracy under controlled conditions	94% accuracy using 128-dimensional facial embeddings
Real-Time Detection	Limited real-time monitoring capability	Continuous real-time CCTV/Webcam surveillance
Alert Mechanism	Requires manual verification and reporting	Automatic email alerts to family and authorities
Database Management	Basic storage of records	Centralized database with case and detection log management
Automation Level	Semi-automated identification process	Fully automated detection, matching, and notification
Response Time	Delayed due to human intervention	Instant detection and notification within seconds

Conclusion and Future Scope

- The proposed AI-based missing person detection system effectively automates face detection, recognition, and alert generation using computer vision techniques.
- The system improves missing person identification accuracy and reduces manual investigation effort compared to traditional methods.
- Real-time surveillance analysis helps authorities identify and track missing persons more efficiently.
- The proposed solution demonstrates faster response time, automated notifications, and efficient case management capabilities.
- Integration of advanced deep learning models can further improve face recognition accuracy under real-world conditions.
- Development of real-time CCTV integration and multi-camera support can enhance large-scale surveillance monitoring.
- Incorporation of geolocation tracking and map-based visualization can improve missing person tracing workflows.
- Deployment as a cloud-based surveillance and alert platform can improve scalability and accessibility for law enforcement agencies.

References

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